

IB MATH AA SL

Lesson 7

Solving Exponential & Log Equations

Number & Algebra · 60-minute lesson

Syllabus SL 1.7 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. Solve $2^x = 16$
2. Solve $\log_3 x = 2$
3. What is $\log 10^x$?

Answers (click to reveal)

1. $x = 4$
2. $x = 9$
3. x

What we're learning today

LEARNING OBJECTIVES

- 1** **Solve**
exponential equations using logarithms.
- 2** **Solve**
logarithmic equations.
- 3** **Apply**
the laws of logs to rearrange.
- 4** **Check**
solutions are valid (log of a positive only).

SUCCESS CRITERIA — I CAN...

- ✓
take logs of both sides to bring down a power
- ✓
use the power law to solve for x
- ✓
rewrite a log equation in index form
- ✓
reject invalid (negative-argument) solutions

Exponential equations

THE METHOD

$$a^x = b \Rightarrow x = \frac{b}{a}$$

Step 1

take logs of both sides

Step 2

use $\log a^x = x \log a$

Step 3

divide to make x the subject

WHY LOGS?

Logs 'bring the power down' so you can solve for x .

GDC

Paper 2: solve directly on the calculator.

Solve an exponential

QUESTION

Solve $3^x = 50$,
giving your answer to 3 s.f.

SOLUTION

Step 1 take logs of both sides:

$$3^x = 50$$

Step 2 power law brings x down:

$$x \log 3 = \log 50$$

Step 3 divide and evaluate:

$$x = \frac{\log 50}{\log 3} = 3.56$$

Your turn — exponentials

ANSWER IN YOUR BOOK

1. Solve $2^x = 20$ (3 s.f.)
2. Solve $5^x = 100$ (3 s.f.)
3. Solve $10^x = 4500$ (3 s.f.)

Answers (click to reveal)

1. $\log 20 / \log 2 = 4.32$
2. $\log 100 / \log 5 = 2.86$
3. $\log 4500 = 3.65$

Logarithmic equations

THE METHOD

$$\square_a x = c \Rightarrow x = a^c$$

Step 1

combine logs into one using the laws

Step 2

rewrite in index form

Step 3

solve, then check x makes every log valid

VALIDITY

You can only take the log of a positive number.

Always check

reject any solution that makes an argument ≤ 0 .

Solve a log equation

QUESTION

Solve $\log_2(x) + \log_2(x - 2) = 3$.

SOLUTION

Step 1 combine with the product law:

$$\log_2(x(x - 2)) = 3$$

Step 2 index form ($2^3 = 8$):

$$x^2 - 2x = 8$$

Step 3 solve $x^2 - 2x - 8 = 0 \rightarrow (x - 4)(x + 2) = 0$;
reject $x = -2$, so $x = 4$

Exam-style question

(a) Solve $4^x = 30$, giving your answer to 3 s.f. [3]

(b) Solve $\log_3(2x + 1) = 2$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $x \log 4 = \log 30$ (M1)(A1) $\rightarrow x = \log 30 / \log 4 = 2.45$ (A1)

(b) $2x+1 = 3^2 = 9$ (M1)(A1) $\rightarrow x = 4$ (A1)

Common mistakes



Forgetting to take logs

You can't solve $a^x = b$ by inspection unless b is a neat power.



Power law

$\log a^x = x \log a$ — bring the power down first.



Not combining

Combine logs into ONE before going to index form.



Invalid solutions

Reject any answer that makes a log's argument ≤ 0 .

Mixed practice

QUESTIONS

1. Solve $2^x = 1000$ (3 s.f.)
2. Solve $\log_5 x = 3$
3. Solve $7^{x+1} = 50$ (3 s.f.)
4. Solve $\log_2 x + \log_2 4 = 5$

Answers (click to reveal)

1. $\log 1000 / \log 2 = 9.97$
2. $x = 125$
3. $x+1 = \log 50 / \log 7 \rightarrow x = 1.01$
4. $\log_2(4x) = 5 \rightarrow 4x = 32 \rightarrow x = 8$

Plenary & exit ticket

KEY FORMULAS

$$a^x = b \Rightarrow x = \frac{\log b}{\log a}$$

$$\square_a x = c \Rightarrow x = a^c$$

EXIT TICKET — before you leave

1. Solve $3^x = 90$ (3 s.f.)

2. Solve $\log_4 x = 2$

3. Solve $10^x = 250$ (3 s.f.)

Answers (click to reveal)

1. $\log 90 / \log 3 = 4.10$

2. $x = 16$

3. $\log 250 = 2.40$

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Exponential & log equations.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 8

The binomial theorem — expanding $(a + b)^n$ quickly.

$$(a + b)^n = \sum \binom{n}{r} a^{n-r} b^r$$